

Cloud Computing - Exam Notes

1. Differentiate between Grid Computing, Distributed Computing, and Cloud Computing

- Grid Computing: Combines resources from multiple locations to solve one large task. Example: SETI@home.
- Distributed Computing: Independent systems connected via network working together. Example: Hadoop cluster.
- Cloud Computing: On-demand IT resources delivered over the internet. Example: AWS, Google Cloud.

Relationship: Distributed $\subset$ Grid $\subset$ Cloud.

2. Characteristics of Cloud Computing Environments

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity and scalability
- Measured service (pay-as-you-go)

3. Barriers of Cloud Computing

- Security & Privacy issues
- Data transfer bottlenecks
- Vendor lock-in
- Downtime risks
- Compliance & legal issues

4. Essential Characteristics of Cloud Computing

Defined by NIST:

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service

5. Big Data Analytics & Role of Cloud

Big Data Analytics: Process of analyzing large datasets for patterns and insights.

Cloud helps:

- Scalable storage
- High processing power (e.g., Spark)
- Cost-effective pay-per-use
- Collaboration and data sharing

6. Service Models (IaaS, PaaS, SaaS)

- IaaS: Virtualized resources. Example: AWS EC2. Use: Hosting apps.
- PaaS: Development platforms. Example: Google App Engine. Use: App development.
- SaaS: Software via internet. Example: Gmail, Salesforce. Use: Business apps.

7. Deployment Models

- Public Cloud: Cheap, scalable but less secure. Example: AWS.
- Private Cloud: Secure but costly. Example: VMware.
- Hybrid Cloud: Combines both, flexible but complex.
- Community Cloud: Shared by similar organizations, collaboration but limited scale.

8. Cloud Computing vs On-Premises IT

Cost: On-premises = High upfront, Cloud = Pay-as-you-go.

Maintenance: On-premises = By org, Cloud = By provider.

Scalability: On-premises = Limited, Cloud = Elastic.

Management: On-premises = Org responsible, Cloud = Provider responsible.

9. Challenges & Risks

- Security & Privacy
- Compliance issues
- Vendor lock-in
- Downtime & reliability
- Hidden costs

10. Cloud Computing Architecture

- Front-end: clients, browsers
- Back-end: servers, storage, virtualization, services
- Middleware: connection layer
- Management: security, billing
- Network: internet & APIs